

Q 1

1. Show that $m(a+bX) = a + b \times m(X)$

$$m(a+bX) = \frac{1}{N} \sum_{i=1}^N (a+bX_i)$$

$$m(a+bX) = \frac{1}{N} \left(\sum_{i=1}^N a + \sum_{i=1}^N bX_i \right) \quad \sum_{i=1}^N a = N \cdot a$$

$$m(a+bX) = \frac{1}{N} \left(N \cdot a + b \sum_{i=1}^N X_i \right)$$

$$m(a+bX) = a + b \left(\frac{1}{N} \sum_{i=1}^N X_i \right) \quad m(X) = \frac{1}{N} \sum_{i=1}^N X_i$$

$$m(a+bX) = a + b \times m(X)$$

2. Show that $\text{cov}(X, a+bY) = b \times \text{cov}(X, Y)$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) ((a+bY_i) - (a+b \times m(Y)))$$

$$(a+bY_i) - (a+b \times m(Y)) = bY_i - b \times m(Y) = b(Y_i - m(Y))$$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (b(Y_i - m(Y)))$$

$$\text{cov}(X, a+bY) = b \cdot \left(\frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (Y_i - m(Y)) \right) = b \times \text{cov}(X, Y)$$

$$\text{cov}(X, a+bY) = b \times \text{cov}(X, Y)$$

3. Show that $\text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$ and in particular that

$$\text{cov}(X, X) = S^2$$

$$\text{cov}(X, X) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (X_i - m(X)) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2 = S^2$$

$$\text{cov}(X, X) = S^2$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N ((a+bX_i) - (a+b \cdot m(X))) ((a+bX_i) - (a+b \cdot m(X)))$$

$$(a+bX_i) - (a+b \cdot m(X)) = b(X_i - m(X))$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N (b(X_i - m(X)))^2 = \frac{1}{N} \sum_{i=1}^N b^2 (X_i - m(X))^2$$

$$\text{cov}(a+bX, a+bX) = b^2 \sum_{i=1}^N (X_i - m(X))^2 = b^2 \text{cov}(X, X)$$

4. Is it always true that a non-decreasing transformation $g(\cdot)$.
Is it always true that $m(g(X)) = g(m(X))$?

No, it's not always true. The reason for this is because the identity $m(g(X)) = g(m(X))$ only holds in general for linear transformations. For non-linear transformations the mean of the transformed data does not equal the transformation applied to the mean.